Sample_Project

Tuffy Licciardi Issa

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This document presents a minimal and reproducible example of my empirical research workflow.

Full code and additional projects are available at: **github.com/tuffyli**

```
set.seed(123)

# Load packages
library(dplyr);library(here);library(tidyverse);library(stargazer);library(MatchIt);library(broom);libra
```

Objective

The objective of this sample is to illustrate an empirical workflow using Brazilian household survey data. I analyze the expansion of alimony rights to women in stable unions and its potential effects on educational attainment among young women.

This policy reform extended alimony eligibility to women in non-civil unions (such as consensual or religious-only unions). The analysis investigates whether this change affected the educational decisions of young women who became newly entitled to this right.

The empirical framework is inspired by Rangel (2006). The main contribution of this exercise is the implementation of an additional Propensity Score Matching (PSM) approach to improve comparability between treated and control individuals.

Data

The analysis uses microdata from Brazil's PNAD household survey.

The sample includes women aged 15–24 observed between 1992 and 1995.

Treated individuals are women in non-civil unions (consensual or religious-only) who are household heads or spouses. The control group consists of otherwise similar women in legally recognized civil unions (civil or civil + religious).

Due to the large size of the original PNAD dataset, the full dataset cannot be distributed. The repository therefore includes a filtered dataset containing only the variables required for the empirical analysis. At **repository**, I include supplementary code detailing the selection of key variables.

Key variables include:

- age
- race/color

- education
- household per capita income
- number of children
- labor market participation
- state of residence

```r
# ------------------- #
# 1. Data Extraction ----

library(here)

data <- readRDS(here("Ava_Pol","data", "final_filtered_9295.rds"))

# 1.1 Summary ----
# In the data manipulation code I created a summary table. Here I present it
summary_tbl <- readRDS(here("Ava_Pol", "data", "summary.rds"))
print(summary_tbl)
```

```
##                                Mean        SD Min     Max
## Treatment                      0.45      0.50   0       1
## Female = 1                     1.00      0.00   1       1
## Race (White or Asian = 1)      0.55      0.50   0       1
## Highest Education              4.22      0.56   0       9
## Age                           21.04      2.41  15      24
## Years of education             6.61      3.37   1      17
## Last grade concluded           4.47      2.00   1       9
## Course enrollment              2.48      1.65   0       9
## Household per capita wage 173221.66 269756.48   0 9549387
## Pension (yes = 1)              0.00      0.05   0       1
## CBO Group                      4.34      2.58   0       8
## Male child (house)             0.57      0.72   0      12
## Female child (house)           0.55      0.71   0       6
## Total childs                   1.17      1.00   0      13
```

Propensity Score Matching

Propensity Score Matching

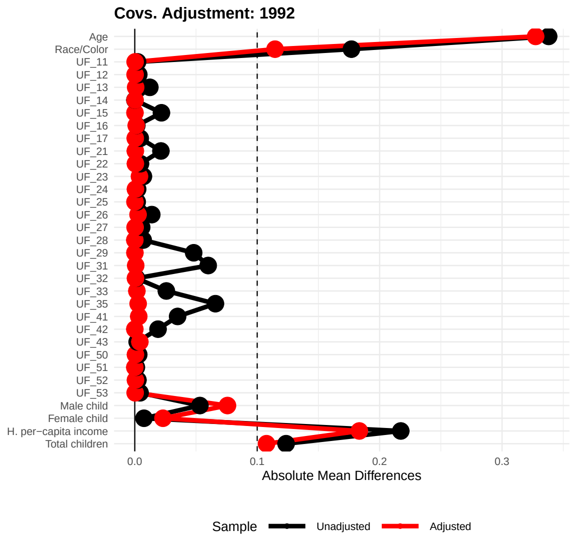
To improve comparability between treated and control individuals, I estimate propensity scores using observable characteristics including age, race, geographic location, household income, and number of children.

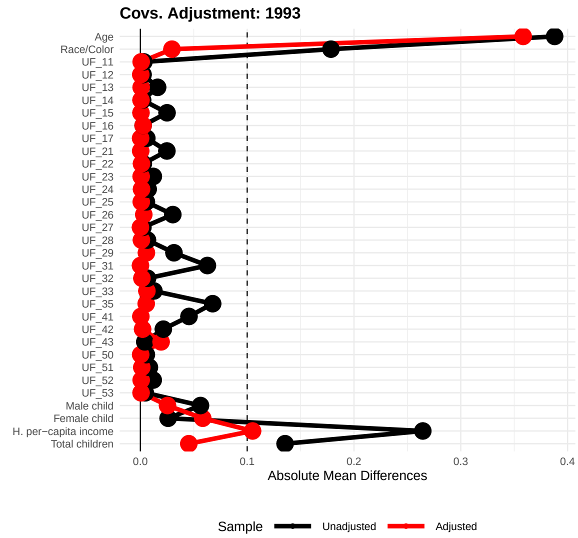
Matching is implemented using nearest-neighbour matching with replacement and a caliper of 0.05.

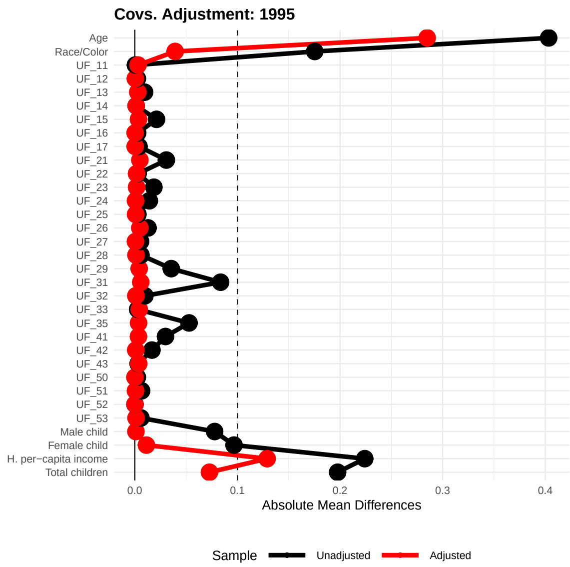
Balance diagnostics are evaluated using standardized mean differences and visualized through love plots. This step allows us to assess whether matching successfully reduces covariate imbalance between treated and control groups before estimating treatment effects.

```r
# ----------------------------------------------------------------------
# 2. PSM balance
# ---------------------------------------------------------------------
plots <- list()

for (year in c(1992, 1993, 1995)) {
  df_psm <- data %>%
    filter(ano == year) %>%
    select(
      treatment,                  #Treatment indicator (1 = treated, 0 = control)
      uf,                         #State of residence
      age,                        #Age
      cor,                        #Color/Race
      anos_estudo,                #Years of education
      peso_pessoa,                #Observation weight
      ultima_serie_concluida,     #Last completed grade
      tipo_curso_frequenta,       #Type of course attended
      grupo_cbo,                  #Occupational group
      trabalhou_ultimo_ano,       #Worked in the last 12 months
      renda_dom_per_capita,       #Household per capita income
      dummy_filhos_homens_dom,    #Children Dummy for living with Father
      dummy_filhos_mulheres_dom,  #Children Dummy for living with Mother
      pensao_dummy,               #Pension dummy
      total_filhos                #Total children
    ) %>%
    mutate(uf = as.factor(uf)) %>%
    filter(
      !is.na(treatment), !is.na(uf), !is.na(cor), !is.na(peso_pessoa),
      !is.na(dummy_filhos_homens_dom), !is.na(dummy_filhos_mulheres_dom),
      !is.na(renda_dom_per_capita), !is.na(total_filhos)
    )

  match_model <- matchit(
    treatment ~ age + cor + uf + dummy_filhos_homens_dom +
      dummy_filhos_mulheres_dom + renda_dom_per_capita + total_filhos,
    data = df_psm,
    s.weights = df_psm$peso_pessoa,
    caliper = 0.05,
    method = "nearest",
    replace = TRUE
  )

  # Show the match summary counts in the document
  sum <- summary(match_model)
  latex_df <- as.data.frame(sum$nn)
  latex_table <- kable(latex_df,
                       format = "latex",
                       booktabs = TRUE,
                       caption = "Estatísticas descritivas ponderadas",
                       align = "lccc")

  #The files are availabre in the repository
  writeLines(latex_table,
```

```r
             paste0("C:/Users/tuffy/Documents/Trabalhos/", year,"_match_summary.tex"))


  # Love plot (INLINE)
  custom_labels <- c(
    distance_weighted         = "Distance (weighted)",
    uf                        = "UF",
    cor                       = "Race/Color",
    dummy_filhos_homens_dom   = "Male child",
    dummy_filhos_mulheres_dom = "Female child",
    age                       = "Age",
    renda_dom_per_capita      = "H. per-capita income",
    total_filhos              = "Total children"
  )

  love <- love.plot(
    match_model,
    stats         = "mean.diffs",
    abs           = TRUE,
    threshold     = 0.10,
    colors        = c("black", "red"),
    line          = TRUE,
    var.names     = custom_labels,
    drop.distance = TRUE
  ) +
    labs(
      title = paste0("Covs. Adjustment: ",year)
    ) +
   theme_minimal(base_size = 5) +
    theme(plot.title = element_text(face = "bold"),
                 legend.position = "bottom")
  #Grouping the plots
  plots[[as.character(year)]] <- love


  matched_data <- match.data(match_model) %>%
    mutate(ano = year)

  assign(paste0("df_matched_", year), matched_data)
  rm(year, custom_labels, sum, ess_table, latex_df, latex_table, love)

}

top    <- cowplot::plot_grid(plots[["1992"]], plots[["1993"]],
                             ncol = 2, align = "hv", rel_widths = c(1, 1))
bottom <- cowplot::plot_grid(plots[["1995"]],
                             ncol = 2, align = "hv", rel_widths = c(1, 1))  # legend stays here

combined <- cowplot::plot_grid(top, bottom, ncol = 1, rel_heights = c(1, 1.05))
combined
```







Regression

To estimate treatment effects, I fit progressively richer regression specifications using the matched sample.

The outcome variable is years of education.

Three specifications are estimated:

1. Baseline model without fixed effects.
2. Model including state-by-year fixed effects to control for geographic shocks.
3. Model including fixed effects and additional individual controls.

All regressions are estimated using matching weights and heteroskedasticity-robust standard errors.

```
# -------------------------------------------------------------------------- #
# 3. Regression -----
```